Team: Netherite

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ANLP Final

Evaluating Multilingual Reasoning through a Chain-of-Thought Lens

Final Presentation

Previous Work + Gap

Previous work has shown:

- Chain-of-Thought (CoT) improves many mathematical and symbolic reasoning tasks (Wei et al.; Sprague et al.).
- Irrelevant sentences can disrupt multi-step reasoning, mainly studied on English arithmetic (Shi et al.; Mirzadeh et al.).
- TRD evaluates multilingual date/time reasoning, but its original study did not evaluate CoT (Mazzia et al.).

Research gap: How do CoT, language of reasoning, and distractors interact on the same English and Hindi temporal problems?

Research Questions (1/2)

RQ1 Does CoT help temporal reasoning, and does that depend on language and difficulty?

RQ2 Does CoT make distractors more harmful?

RQ3 How much of the English-Hindi gap follows the language of reasoning rather than the question language?

Research Questions (2/2)

RQ4 Does asking for Hinglish reasoning help?

RQ5 Does divergence between English and Hindi reasoning traces track whether their answers disagree?

RQ6 Can interventions show that distractor mentions in CoT traces cause errors?

Dataset: Temporal Reasoning Dataset (TRD)

6,750 base questions per language

9 tasks	date/time addition, subtraction, duration; recurrence; day of week; interval date
5 difficulty levels	short to very very long
150 questions / task / level	matching question IDs in English and Hindi
3 input variants	no distractor, time-related distractor, unrelated distractor

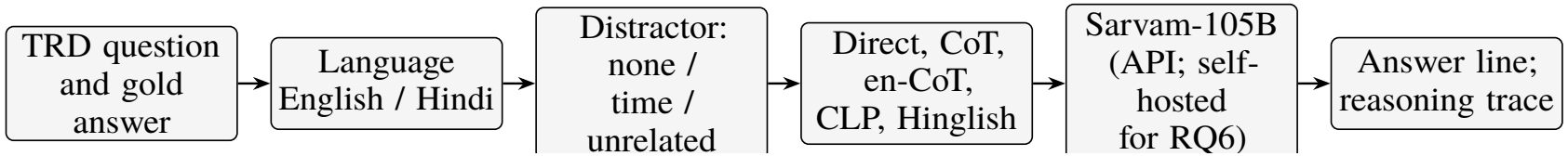
All 13,500 language-specific gold answers were rechecked with Python datetime.

Prompting Conditions

Direct	answer only; no written reasoning
CoT	reason step by step in the question's language
en-CoT	Hindi question, reasoning requested in English
CLP	two turns: understand Hindi in English, then solve
Hinglish	mix English and Hindi in Roman script, tested on both question languages

Direct and CoT: all 6,750 questions. Additional prompting arms: 2,250-question subset (50 per task/level).

Experiment Pipeline



Controlled generation

- Sarvam-105B; temperature 0; thinking off; 4,096-token budget.
- Fixed answer line enables automatic accuracy scoring.
- Responses also provide reasoning traces for operation, language, and distractor analyses.

Evaluation + Reliability

- Answered accuracy excludes responses with no final answer line; strict accuracy treats these as wrong.
- 95% confidence intervals use question-clustered bootstrap, accounting for three distractor variants of each question.
- Task tests use McNemar with Holm correction; trace analyses use clustered or stratified comparisons.
- The operation parser captures 95-96% of standard CoT segments, but automatic labels and trace metrics are imperfect.

Keep answered vs strict accuracy separate when cut-offs change the interpretation.

Results

Final Results: Six Short Answers

RQ1 CoT gain: English +29.5 pts; Hindi +4.7 pts.

RQ2 Hindi distractor penalty: +2.2 pts, largely one sentence.

RQ3 Reasoning language accounts for 23.5 of the 29.5-point gap.

RQ4 Hinglish: -18.0 pts on English Q; essentially no change on Hindi Q.

RQ5 Higher trace divergence is associated with disagreement: OR 4.93 per 0.1 RDS.

RQ6 Removing distractors helps slightly; trace mentions are not proven causal.

RQ1: CoT Helps English Far More Than Hindi

Language	Direct	CoT	Gain
English	59.0%	88.5%	**+29.5 pts**
Hindi	54.7%	59.4%	**+4.7 pts**

English-Hindi gap: 4.2 points with direct answering -> 29.1 points with CoT.

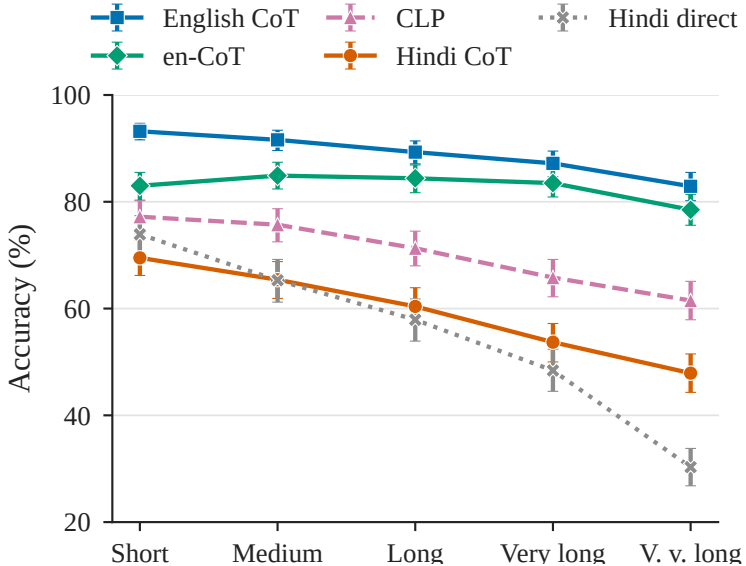
95% CIs on CoT gains: English [28.5, 30.6], Hindi [3.6, 5.7] points.

RQ1: Some Hindi Tasks Get Worse with CoT

Task	English direct -> CoT	Hindi direct -> CoT
Date subtraction	86.0 -> 95.2	84.8 -> 66.0
Interval date	43.2 -> 50.7	46.8 -> 23.9
Date recurrence	22.1 -> 90.3	2.8 -> 20.1
Day of week	14.9 -> 69.5	15.2 -> 21.0

The two Hindi declines remain significant in the full 6,750-question study: -19.9 pts and -23.3 pts.

RQ1: Performance by Difficulty



Full-set CoT gap: 24.0 to 34.9 pts from short to longest; trend p = 0.21 (not significant).

RQ2: Average Distractor Harm Is Small

Hindi CoT, time-related distractor

-2.6 pts accuracy compared with no distractor

- Additional drop versus Hindi direct answering: +2.2 points (CoT penalty).
- The interval reaches zero when the 45 task-sentence units, rather than questions, are sampled.
- This revises the midterm claim: distractors do not robustly make CoT fragile only in Hindi.

RQ2: One Distractor Dominates

"The last time I had a weekly commitment was in January."

- On 134 date-recurrence questions, strict English CoT falls 88.8% -> 19.4%; cut-offs rise from 4 to 92.
- Hindi CoT accuracy on those questions falls 26.9% -> 2.2%.
- Time-related distractors appear in 25.0% of Hindi traces vs 8.5% of English traces; Hindi mention-correctness OR = 0.77.
- Caution: "January" competes with the start date, so the sentence is not purely irrelevant.

RQ3: Most of the Gap Follows Reasoning Language

English Q + English CoT

88.9%

Hindi Q + English en-CoT

82.9%

Hindi Q + Hindi CoT

59.4%

6.0 pts

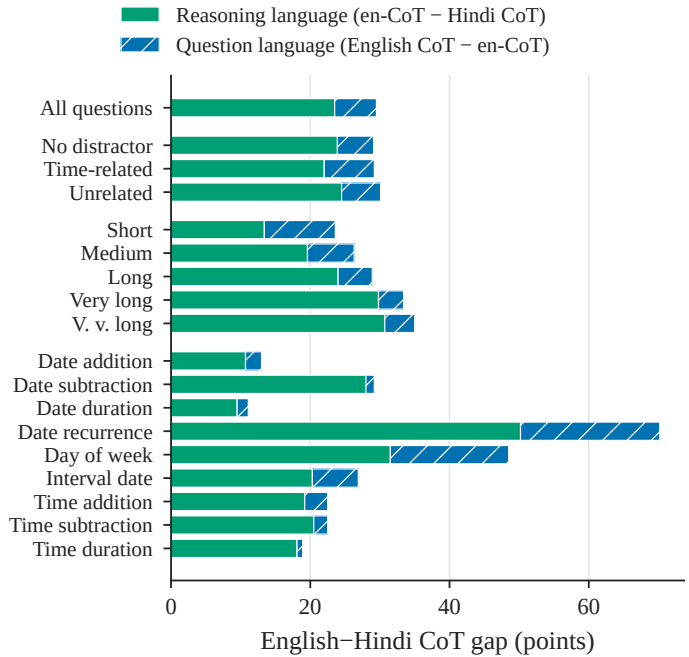
question language

23.5 pts

reasoning language

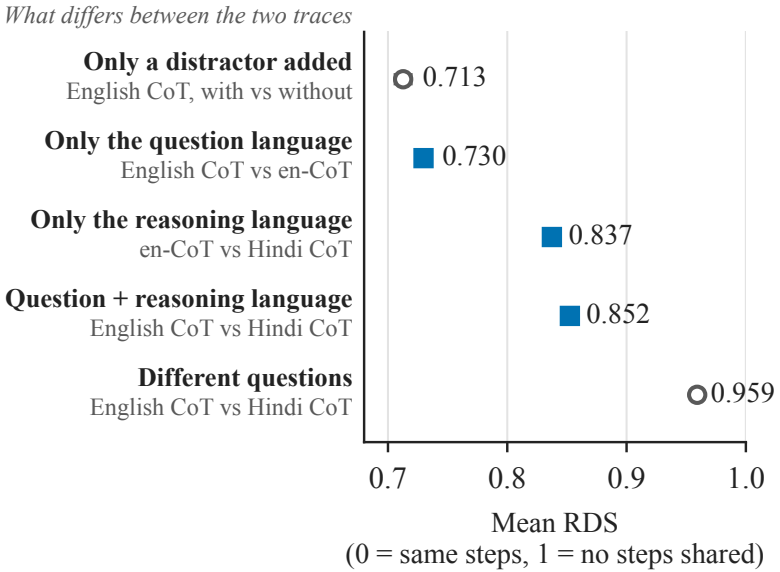
Reasoning language represents 79.7% of the 29.5-point gap, along this measured comparison path.

RQ3: Breakdown Across Tasks + Conditions



Across every task and distractor type, the reasoning-language component is larger than the question-language component.

RQ3: Reasoning Traces Also Split by Language



Verification phrases

English CoT: 86.1%

en-CoT: 74.2%

Hindi CoT: 1.3%

Similar question != similar traces when reasoning language changes.

RQ3: CLP + Thinking-Mode Check

Cross-lingual prompting (CLP)

- CLP: 70.3% on Hindi questions; +10.9 points versus Hindi CoT, but -12.6 versus en-CoT.
- CLP understanding turn often solves the task already; final turn frequently copies its answer (observational).

Thinking-mode pilot

- Hindi CoT reaches 97.9% answered (89.9% strict).
- Hidden reasoning is 79% English words; token budget is four times larger, so effects cannot be isolated.

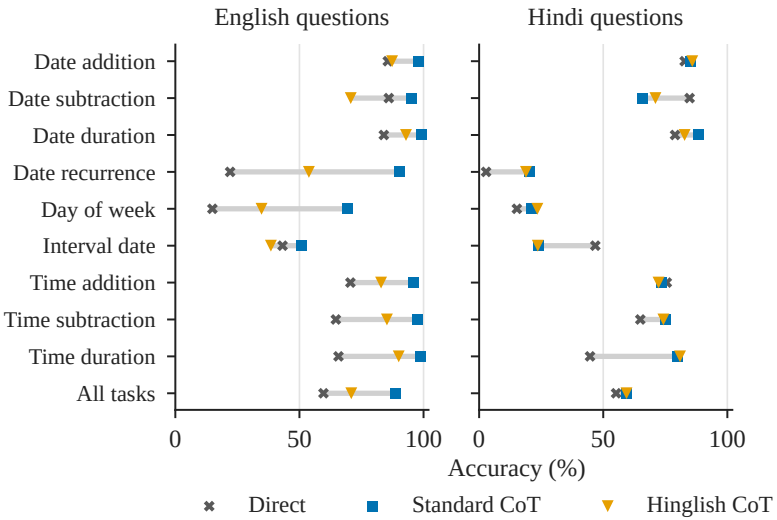
RQ4: Hinglish Does Not Help

English Q: -18.0 pts

Hindi Q: ~0.0 pts

On Hindi questions, 68.5% of Hinglish traces are mostly in Devanagari, despite the Roman-script instruction.

English verification frequency falls from 86.5% to 18.6%.

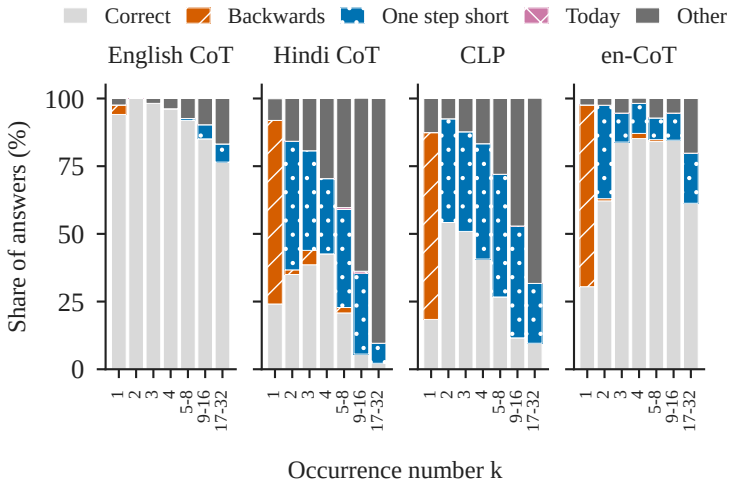


Error Analysis: The Model Uses Different Methods

Error pattern	Hindi CoT	English CoT
Date subtraction: answer on/after today	**13%**	**0%**
Interval date: span not seven days	**21%**	**6%**
Date recurrence: one interval short	**35%**	**21%**
Day of week: exactly one day late	**-**	**58%**

Percentages are shares of wrong, answer-formatted responses for that task; errors were auto-labelled, not hand-checked.

Error Analysis: Reading Hindi Remains Hard



On Hindi recurrence questions with k = 1, around two-thirds of responses count backwards, even with English reasoning:

en-CoT 67.1%

Hindi CoT 67.8%

CLP 69.0%

English CoT: 3.5% backward for k = 1. Changing reasoning language does not fix Hindi question parsing.

RQ5: Can We Compare Reasoning Traces?

Reasoning Divergence Score (RDS)

Parse intermediate operations with operands (e.g., SUB 2025-05-26 4), then calculate normalized edit distance between English and Hindi operation sequences.

0 = same steps in same order

1 = no steps in common

Important caveat: the extracted sequences contain final-answer tokens, so different answers inflate RDS even if the reasoning method is similar.

RQ5: Logistic Regression - Experiment Setup

- Pair English and Hindi CoT traces for the same question, distractor condition and thinking mode.
- Define disagreement = 1 if exactly one language gets the answer correct; otherwise 0.
- Predict disagreement from the Reasoning Divergence Score (RDS) using logistic regression.
- Control for task, difficulty, distractor type and thinking mode. Cluster standard errors by question.
- Second model also controls for normalized trace-length gap to test a possible edit-distance confound.

20,662 paired traces | 19,471 with thinking off

RQ5: Divergence Predicts Disagreement

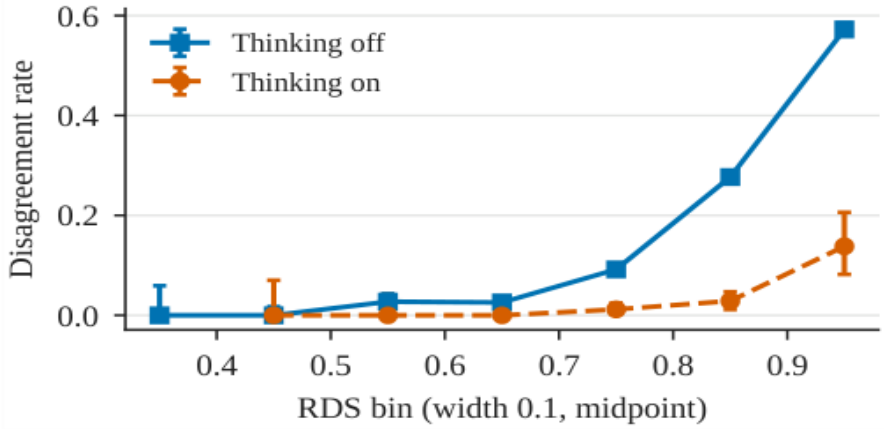


Figure 7: Observed English-Hindi disagreement rate by RDS bin of width 0.1, unadjusted, with 95% intervals; bins with fewer than 20 pairs are hidden.

31.4%
of pairs disagree

0.917 vs 0.809
mean RDS: disagree vs agree

OR 4.93
per +0.1 RDS, adjusted

OR 5.58
with length-gap control

Association, not causation; RDS and correctness come from the same model generations.

RQ6: Intervene on Distractor Mentions

Selected originally wrong traces that mention a time-related distractor: 526 Hindi, 37 English. Find the first mentioning sentence k.

P	keep prefix before k
Q	same prefix; remove distractor from question
M	keep prefix including sentence k
P1	stop one sentence before P

Experiment 1 uses greedy deletion vs neutral deletion; Experiment 2 samples 7 continuations at temperature 0.8.

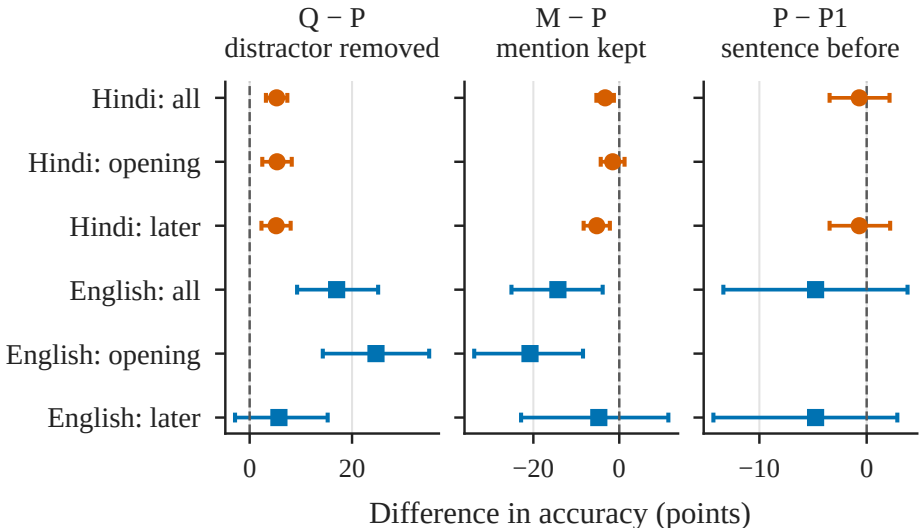
RQ6: Deleting the Written Mention

+0.0 points

over deleting a neutral sentence

- On 222 Hindi traces, deleting the distractor mention did not outperform deleting a length-matched neutral sentence.
- 95% interval: [-6.8, 7.2] points.
- One continuation per variant gave only ~24% power to detect a 5-point effect: no detectable difference is not proof of no effect.

RQ6: Resampling Helps, but Does Not Prove Cause



Hindi Q - P

+5.3 pts

95% CI [3.2, 7.4]

The January sentence supplies 58% of the estimated effect.

The written mention is not distinguishable from an ordinary preceding step; all experiments start from wrong traces.

Main Findings

- CoT is not language-neutral: approximately +30 points in English but only +5 in Hindi.
- Most of the CoT gap follows the language used to write reasoning, rather than the question language.
- Hinglish does not improve Hindi accuracy; one English reasoning turn outperforms two-turn CLP.
- English-Hindi trace divergence is strongly associated with correctness disagreement (adjusted OR 4.93 per 0.1 RDS), but is not causal evidence.
- Distractor mentions do not prove they caused the model to fail; interventions remain inconclusive.

Limitations

- One model, two languages, one dataset seed, and one prompt per condition; run dates overlap with API changes.
- The 2,250-question subset and fixed distractor pool limit precision and generalization to new sentences.
- Hindi distractor translations and some question templates are not strictly parallel or natural.
- Regex-based verification/mention detection and automatic error labels are not human-validated.
- Cross-condition comparisons change both instructions and language; the RQ6 self-hosted model differs from API runs.

Future Work

- Try prompt paraphrases and English CoT without a forced reasoning-language lock.
- Test CLP with solving prohibited during the understanding stage.
- Compare thinking on and off using equal token budgets.
- Run question-by-mention factorial interventions on both correct and wrong traces.
- Validate regex parsing and error labels through human annotation.

Thank You

Questions?

Team Netherite